

ASTRA-ERA RESEARCH EDITION

Do Slugfester Rankings Measure Stable Performance?

Why the broad ordering is informative, exact places are fragile, and a larger sample changes what a ranking means.

0.86

median rank correlation
between random halves

0.17

median score gap between
adjacent displayed ranks

50

ranked speakers meeting
the three-appearance minimum

The leaderboard contains a repeatable signal, but its precision is easy to overread. Typical 95% rank ranges span 12 positions under observed-score resampling and 19 under a pooled-variation model.

PHIL STILWELL · SLUGFESTER

September 4, 2026 | Frozen archive: 253 assessments

New analysis and writing; not an Astra re-judging of the source debates.

Executive summary

The leaderboard contains a repeatable broad pattern, but its exact places are much less certain than the displayed ordering suggests. In the September 4 archive, 50 speakers meet the three-appearance minimum. Their 334 eligible appearances show substantial persistent differences between speaker averages. Among the 23 speakers with at least six appearances, repeatedly dividing their records into two random halves produces a median rank correlation of **0.86**: people who rank relatively high in one half usually rank relatively high in the other.

That broad stability does not justify reading every neighboring place as a meaningful distinction. The median difference between adjacent displayed means is only **0.17 score points**. Under observed-score resampling, the median 95% rank range spans **12 positions**. A model that allows ordinary performance variation even when a person's observed sample is small gives a median span of **19 positions**. These ranges answer somewhat different questions, but both warn against overinterpreting exact rank.

Sample size changes the meaning of an average. Sean Carroll's mean is 86.00 from three appearances; Matt Dillahunty's is 85.93 from 28. The 0.07-point difference is not credible evidence that Carroll is a stronger debater. In the pooled-variation model, Carroll's rank range is 1-17, while Dillahunty's is 2-9. More observations do not guarantee a higher score; they usually make the score a better-supported description of the sampled performance.

The model-based estimate of single-appearance repeatability is **0.63**. Under its assumptions, averaging three appearances raises repeatability to about **0.84**, five to 0.89, and ten to 0.94. These are not probabilities that the AI is correct. Stable scores can reflect stable judging preferences, recurring topics, or opponent selection as well as persistent argumentative skill.

The full ranking analysis covers **237 comparable debates, 474 appearances, and 158 speakers**. Seventy-six speakers appear only once and 32 twice; they are not in the default ranked field. The leaderboard therefore describes a selected repeat-appearance sample, not the world's best thinkers or all participants on the site.

The recommendation is to **retain the ranking as a guide, while reading it through appearance counts, uncertainty ranges, and source context**. Broad bands are informative. Exact places, especially in tightly packed regions or small samples, remain provisional.

Reader's guide

Executive summary	2
1. What should a ranking mean?	4
2. Who enters the leaderboard - and who does not?	4
3. Broad differences are larger than ordinary within-person variation	5
4. Why averaging helps - and what it cannot fix	6
5. A direct check: split each person's record in two	7
6. Exact rank is a more demanding claim	7
7. Why small samples can make resampling overconfident	8
8. More appearances help, but the density of the field also matters	10
9. The top ten are not equally secure	11
10. Assessment process and opponent context	12
11. What an honest leaderboard should communicate	12
12. What the next review should test	13
13. Conclusion: preserve the signal, retire the false precision	13
Appendix: the complete 50-speaker field	14
Methods and sources	16

Reading the evidence

Start with the summary and figures for the main argument. The worked examples explain the statistical ideas; the final methods and sources show how the numbers can be checked. Counts and results refer to the frozen September 4 archive, not to later additions to the site.

1. What should a ranking mean?

A useful ranking should tell us more than who happened to receive a high score once. It should summarize a pattern that is likely to recur under reasonably similar conditions. At the same time, it should not pretend that a small difference between two averages reveals a deep difference between two people.

SLUGFESTER scores argumentative performance in selected debate transcripts. The public speaker average is therefore an average of those assessed performances. It is not a measure of intelligence, moral worth, academic expertise, or the truth of a person's worldview. A brilliant researcher can argue poorly in a particular exchange; a speaker with a false conclusion can still display care in the reasoning they present.

The statistical question is whether the recorded performances contain a persistent speaker-associated pattern, and how precisely the available data locate each speaker within the field. Those are related but separate questions. A field can have a stable broad order while many individual places remain interchangeable within the available uncertainty.

This paper treats September 4, 2026 as a frozen reference point. Its 253-assessment archive includes 237 comparable locked one-on-one assessments. The underlying scores are labeled 5.6 Sol. The Astra-era work reanalyzes the rankings; it does not provide an independent new-model reassessment of each speaker.

2. Who enters the leaderboard - and who does not?

The analysis includes one appearance for each side of each eligible one-on-one debate: **474 appearances by 158 speakers**. Speaker names are resolved using the site's canonical identity records so that variants such as Joe and Joseph Schmid do not become separate competitors. Multi-speaker formats are not silently treated as equivalent individual performances.

The default three-appearance minimum leaves **50 ranked speakers and 334 appearances**. The other 108 speakers have only one or two eligible appearances. Their absence from the default field is a sample-size rule, not a conclusion that they are weaker than the ranked participants.

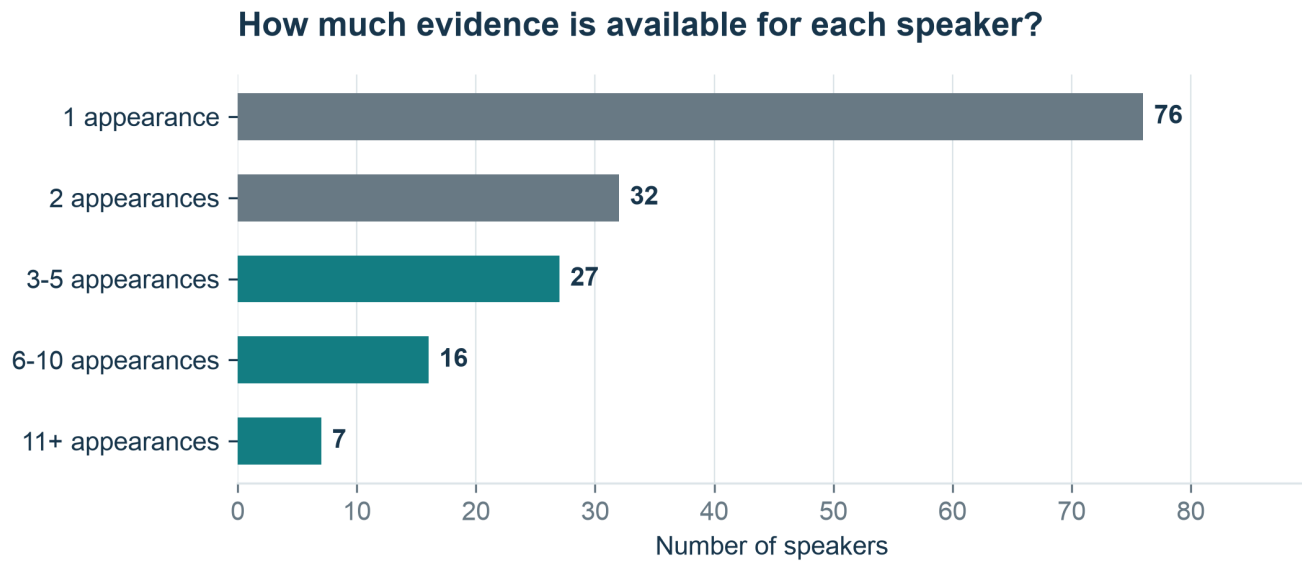


Figure 1. Most speakers have very limited coverage. The three-appearance threshold excludes 108 of the 158 eligible speakers from the default ranked field. A missing rank therefore often means insufficient repeated evidence, not poor performance.

Scope: 158 speakers across 474 eligible appearances; minimum three for rankings.

Seventy-six speakers appear once, 32 twice, 27 three to five times, 16 six to ten times, and seven at least 11 times. The uneven coverage means that the word “average” carries very different evidential weight across the field. An average based on three narrow discussions is not as broad a performance sample as an average based on 28.

The minimum of three is defensible as a simple publication rule, but it is not a magical threshold at which a stable ability suddenly becomes measurable. A three-debate sample can still be unusually favorable or unfavorable. A larger sample can still be biased if all appearances cover one specialty or face similar opponents.

Raising the minimum would increase the amount of evidence per ranked person while excluding more of the field. Lowering it would make the table more inclusive while increasing uncertainty. Neither choice removes the need to show counts and explain what the ranking summarizes.

3. Broad differences are larger than ordinary within-person variation

The repeatability calculation separates two kinds of variation: differences between speaker averages and fluctuations among the appearances of the same speaker. It uses a simple model in which a person's observed score combines a persistent speaker-associated level with ordinary appearance-to-appearance variation.

The estimated variance of the persistent component is **15.81**, and the within-speaker variance is **9.33**. Variance expresses squared deviations, so those numbers are less intuitive than their square roots. The within-speaker standard deviation is about **3.05 score points**: a useful indication of the typical spread of a speaker's observed performances around their own mean in this model.

The ratio $15.81 / (15.81 + 9.33)$ is approximately **0.63**. This is the estimated single-appearance repeatability. It says that a substantial part of the modeled variation is associated with which speaker is being assessed rather than which of that speaker's appearances happened to be sampled.

It does not tell us why the persistent component exists. It may include genuine differences in reasoning, but also stable topic expertise, recurring argumentative roles, opponent selection, recognizable presentation styles, or systematic preferences in the judging process. A repeatable measurement can be biased. Reliability is not the same thing as truth or fairness.

4. Why averaging helps - and what it cannot fix

If appearance-to-appearance fluctuations are approximately independent, averaging reduces their contribution. In the fitted model, a three-appearance mean has repeatability around **0.84**; a five-appearance mean around **0.89**; and a ten-appearance mean around **0.94**. The model does not predict that a speaker will improve as they appear more often. It predicts that their average will be less affected by ordinary variation.

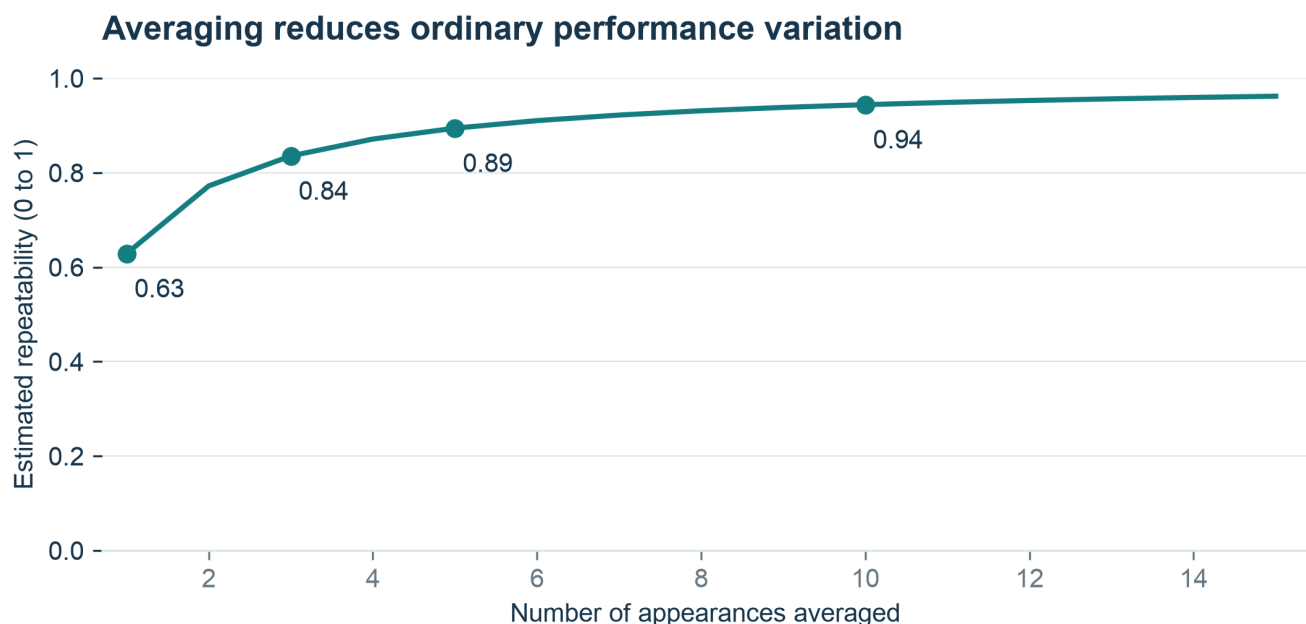


Figure 2. The curve is a conditional projection from the fitted repeatability model, not a newly observed experiment at each sample size. It assumes comparable appearances with approximately independent ordinary variation.

Scope: One-way repeatability model fitted to 334 appearances by 50 speakers; conditional projection.

An intuitive approximation uses the square-root rule. If the ordinary spread is 3.05 points, the uncertainty in a mean shrinks roughly in proportion to 1 divided by the square root of the number of appearances. Three appearances therefore leave much more ordinary variation in the average than ten or 28. Doubling the sample does not halve uncertainty; reducing it by half typically requires about four times as many comparable observations.

That rule is helpful, but it is conditional. If all new appearances come from the same narrow topic or share a systematic judging preference, they may add less independent information than their count suggests. Repeating a biased procedure can make an estimate more stable without making it more accurate.

More data can narrow uncertainty without correcting bias. A scale that consistently reads two kilograms high becomes no more accurate because a person steps on it 20 times. The average is more repeatable, but the offset remains. Debate rankings need both repeated evidence and scrutiny of the scoring instrument.

This is why paper six's process audit matters to the leaderboard. Absolute score levels differ between the earlier and later assessment families. A speaker's average can be affected by how many appearances come from each. Large samples help with ordinary fluctuation, not automatically with that comparability problem.

5. A direct check: split each person's record in two

The repeatability model is not the only evidence. For the **23 speakers with at least six appearances**, the analysis repeatedly divides each person's scores into two random halves, computes a mean from each half, and compares the resulting rankings. If the leaderboard were mostly random fluctuation, high placement in one half would reveal little about placement in the other.

Across 3,000 random splits, the median **Spearman rank correlation** is **0.86**. Spearman correlation compares relative ordering rather than requiring the numerical averages to match exactly. The middle 95% of the split results runs approximately from 0.79 to 0.92. This is substantial broad agreement.

Those limits describe variation across the chosen splits, not a universal confidence interval for how well the leaderboard predicts future performance. The same recorded scores are reused in different divisions. The exercise tests internal repeatability within the available archive, not external validity on a new, independently selected set of debates.

A second check divides those speakers' appearances by archive sequence: earlier-numbered versus later-numbered records. Its rank correlation is **0.84**. This is a useful harder separation than a purely random split, but it is not a test of chronological improvement or decline. The archive sequence is not the filming date, and different assessment processes appear unevenly across it.

Taken together, the model and split checks support a persistent broad signal. They do not establish that the displayed rank of every individual is stable, or that the signal isolates a general ability independent of topics and judging conventions.

6. Exact rank is a more demanding claim

The median difference between adjacent displayed speaker means is **0.17 points**. Many neighbors are separated by fractions of a point, while ordinary within-speaker variation is measured in several whole points. This does not make the ordering arithmetically wrong. It makes the substantive distinction between adjacent places easy to overread.

The table orders Joseph Schmid first at **87.33 from six appearances**, Alex Malpass second at **87.17 from six**, and Scott Clifton third at **86.75 from four**. Those are accurate descriptive means for the frozen sample. They should not be translated into strong claims that the first person would reliably outperform the second or third in a matched new debate.

To examine rank sensitivity, the analysis repeatedly resamples each speaker's observed scores, keeping that speaker's sample size fixed, and recalculates the 50-person field. The middle 95% of simulated ranks becomes an **observed-score resampling range**. Tied simulated means receive average ranks, so alphabetical tie-breaking does not create artificial certainty.

The median width is **12 rank positions**. A supplementary calculation gives each whole debate a shared positive random weight, so both opposing appearances rise or fall in influence together. That paired-weight check also gives a median width of 12. The precise endpoints vary, but the warning about exact places is not removed by keeping opponents paired.

7. Why small samples can make resampling overconfident

Observed-score resampling can only reuse what has been seen. If a person has three similar scores, every replacement sample will contain those same few values. The resulting mean and rank ranges may look narrow even though a fourth genuinely new performance could differ substantially.

A second model addresses that limitation by allowing a level of ordinary performance variation estimated from the whole eligible field. It combines each person's observed mean and sample size with the shared within-speaker variation. Sparse samples are pulled somewhat toward the field's typical level; larger samples remain closer to their recorded means. This is often called partial pooling, but the everyday idea is simply **using the broader record to avoid treating three appearances as complete knowledge of a person**.

The model produces a median 95% rank-range width of **19 positions**, wider than the observed-score resampling result. It does not prove that 19 is the correct amount of uncertainty. It rests on assumptions about a roughly bell-shaped distribution of persistent levels and similar ordinary variation across speakers. It also treats fitted variance estimates as fixed, so it does not include every layer of uncertainty.

Current leaders and their plausible rank ranges

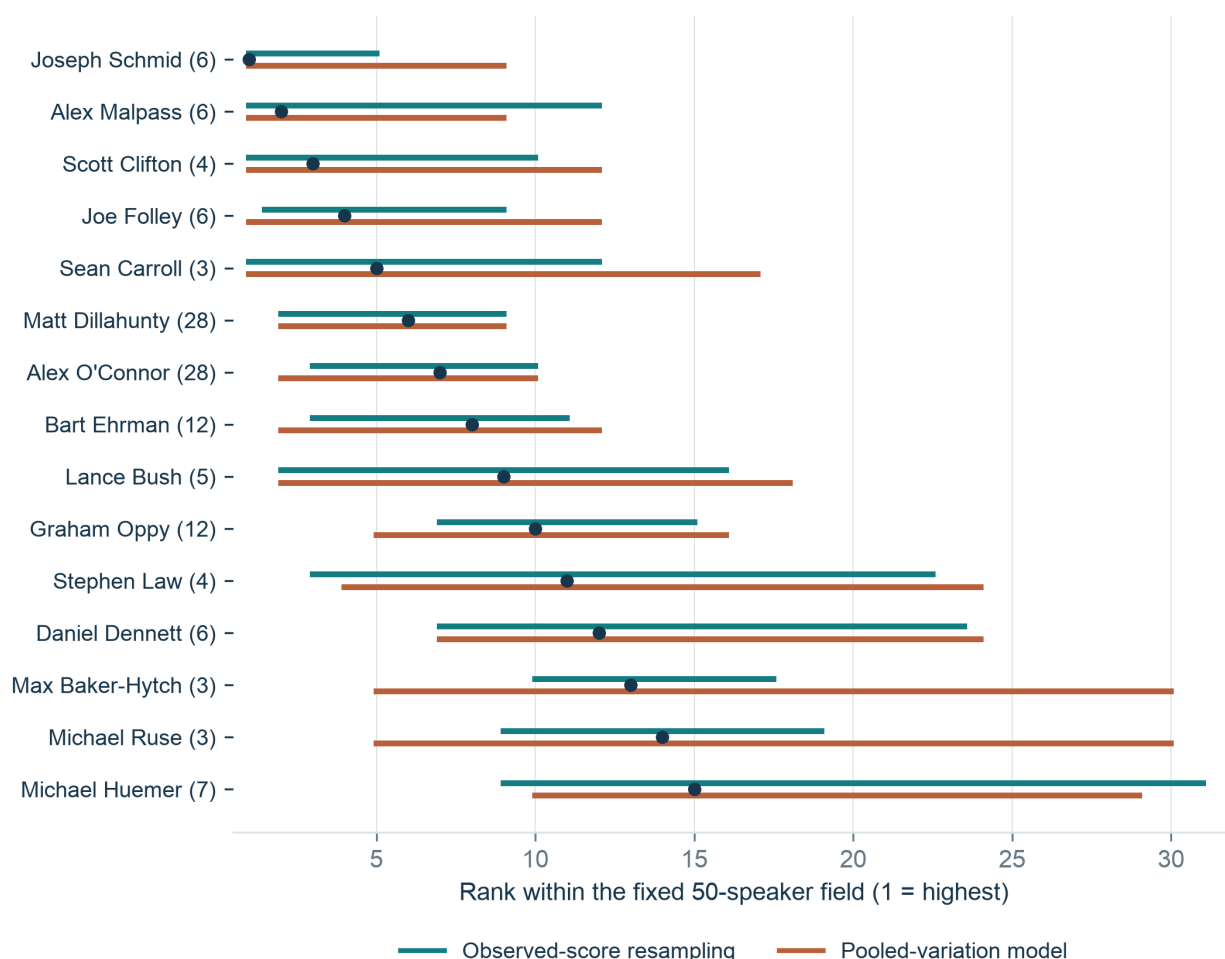


Figure 3. Dots mark the current displayed order for the top 15 mean scores. Teal ranges resample observed scores; rust ranges use a pooled-variation model. The methods answer different conditional questions, so agreement about broad uncertainty is more important than treating either set of endpoints as exact.

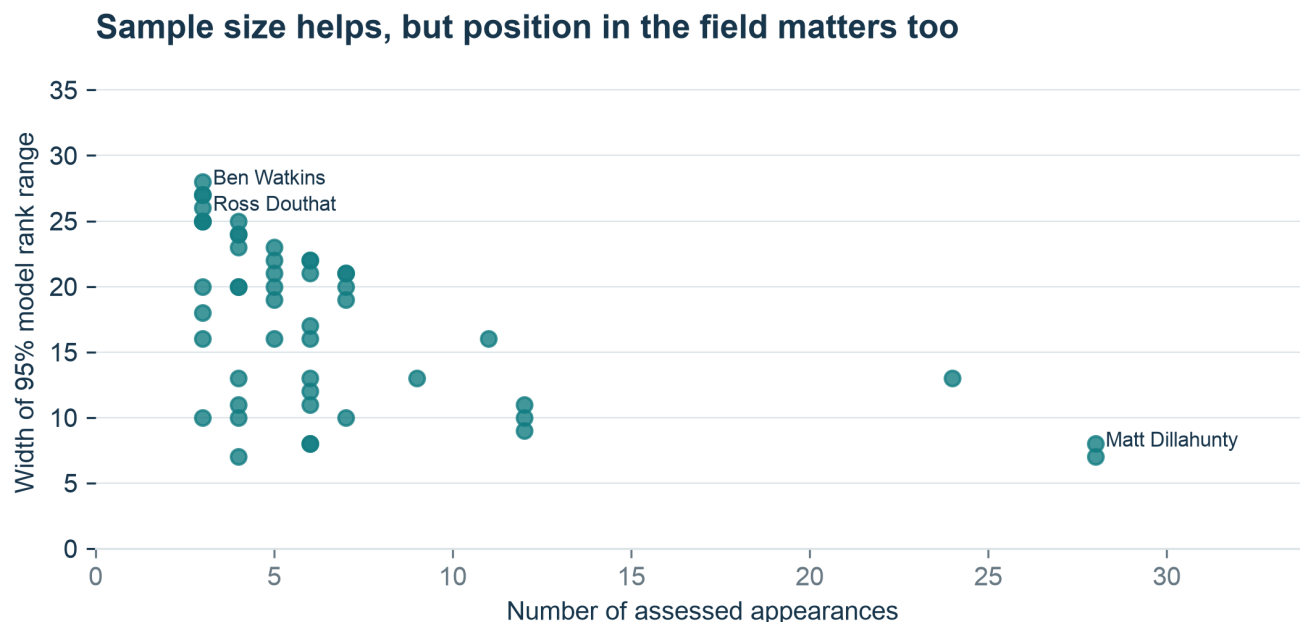
Scope: Top 15 current means; 95% ranges from different uncertainty models; dots show display order.

The comparison between Sean Carroll and Matt Dillahunty is especially instructive. Carroll's mean is **86.00 from three appearances**; Dillahunty's is **85.93 from 28**. The raw gap is only 0.07 points. Their model ranges are 1-17 and 2-9, respectively. The apparent advantage in the displayed means is much smaller than the uncertainty surrounding it.

The point is not that small samples should be penalized as bad performances. Their observed means remain what they are. The issue is how much confidence to place in their location relative to everyone else. A ranking can preserve the descriptive average while showing that its implied precision is limited.

8. More appearances help, but the density of the field also matters

Rank uncertainty depends on two things: how uncertain a speaker's mean is and how tightly packed the surrounding means are. A one-point change near a crowded middle can move a speaker many places. The same change near an isolated end of the field may move them very little.



9. The top ten are not equally secure

The pooled-variation model also estimates how often each speaker appears in the top ten across 20,000 draws. These are **model-conditional inclusion probabilities within the fixed 50-speaker field**, not probabilities that a person is objectively among the world's ten best debaters.

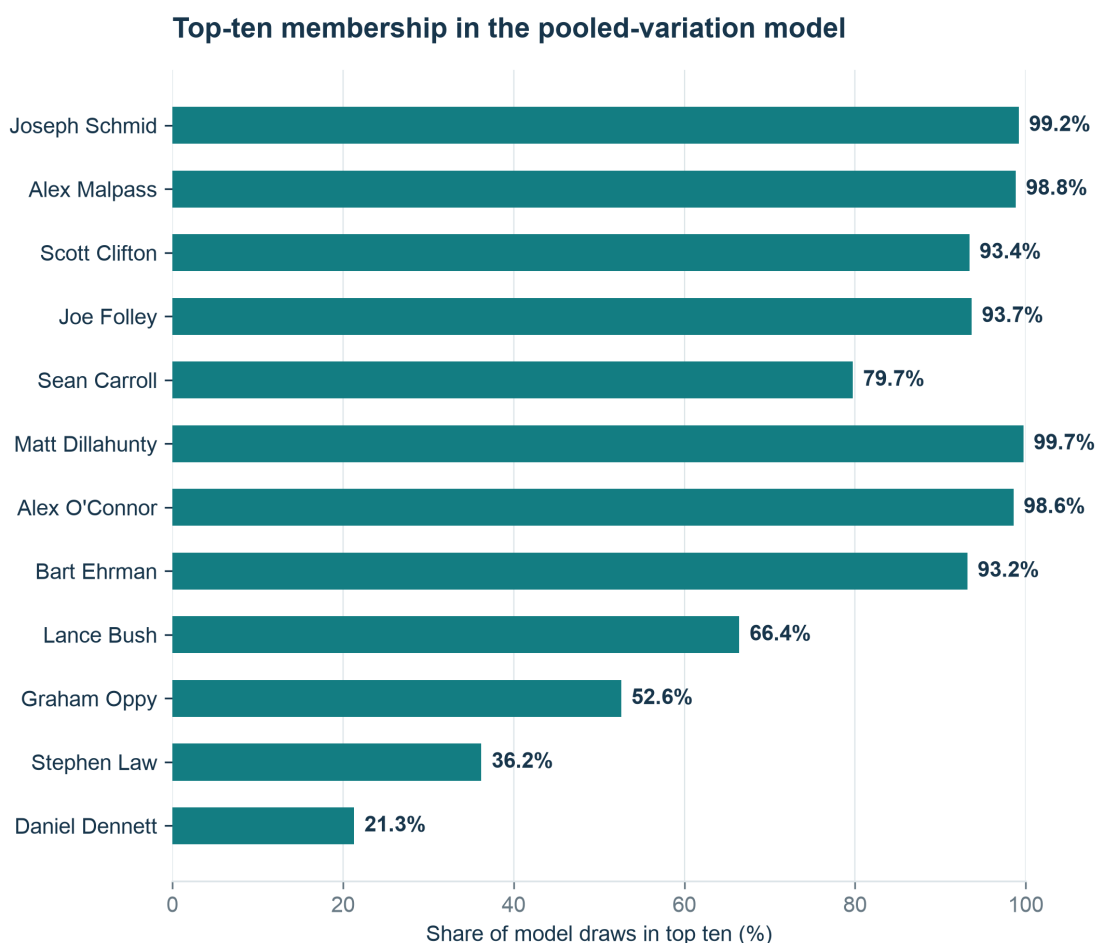


Figure 5. Some current top-ten positions are far more secure under the model than others. High inclusion is conditional on the recorded field, its selected debates, the scoring system, and the model assumptions.

Scope: 20,000 draws of latent means; fixed 50-speaker field; model-conditional probabilities.

Seven of the current ten have top-ten inclusion above 90% under this model. Sean Carroll's is about 79.7%, Lance Bush's 66.4%, and Graham Oppy's 52.6%. Stephen Law, currently just outside the top ten, has about 36.2% inclusion. The boundary is therefore a zone of uncertainty, not a sharp divide between two established classes of performer.

Such probabilities are often more useful than a claim that a speaker is “ranked tenth.” They show that a boundary can be unstable even when the broad set of leading performers is reasonably persistent. But they should never be detached from the assumptions. A different selection of debates, scoring process, or model of variation could change them.

The result also depends on keeping the field fixed. Adding a previously unranked speaker with several strong performances can move everyone else down without any of their own scores changing. Rank is relational: it

describes a position among the current competitors, not a property carried by a person independently of the field.

10. Assessment process and opponent context

Paper six finds that later-process score levels are lower on average. To test whether that alone reorganizes the leaderboard, we subtract each process family's mean from the appearances produced by that family and recompute speaker ordering. The correlation between the original and centered ranks is **0.99**. The broad ordering survives this simple adjustment.

Individual changes still matter. Christopher Hitchens moves from displayed place 17 to centered place 22; Richard Dawkins from 19 to 14; Ben Watkins from 24 to 19; and James White from 32 to 27. These movements are sensitivity results, not a corrected official ranking. Subtracting a family mean cannot establish what the same debate would have scored under another process.

Opponent context is another unresolved influence. Some speakers repeatedly face highly experienced critics; others appear against a narrower or less demanding set of opponents. A score is intended to reflect the argument's performance, not simply whether the opponent was strong, but opponent quality affects which objections arise and how much argumentative work is required.

The current analysis does not supply a validated opponent-adjusted ranking. With many sparse speaker records and a selected debate network, a complicated adjustment could appear precise while depending heavily on assumptions. The safer conclusion is to make context inspectable and reserve stronger claims for matched comparisons or better-connected samples.

Topic specialization matters too. A speaker repeatedly debating their strongest subject may have a higher mean than someone taking on a wider range of difficult motions. Neither average is false; they summarize different portfolios. A general ability ranking would need more deliberate balancing than this archive currently supplies.

11. What an honest leaderboard should communicate

The observed average should remain visible, along with the number of eligible appearances. Readers should be able to distinguish a three-debate result from a 28-debate result immediately. A displayed order can be convenient for browsing, but the surrounding explanation should make clear that nearby places are often not meaningfully separated.

Broad performance bands would better reflect the evidence than treating every ordinal place as a distinct achievement. Such bands should not be invented from arbitrary cutoffs without testing. One practical starting point is to show uncertainty ranges and let readers see overlapping evidence rather than presenting a false categorical separation.

The site should also make each speaker's underlying debate list and process exposure easy to inspect. A user ought to be able to ask whether an average comes from many topics, one favored subject, one opponent, or several assessment families. Those are substantive questions, not merely statistical housekeeping.

Changes over time should preserve the reason for the change. A rank can move because a speaker receives a new assessment, because competitors enter the field, because a source correction changes an old score, or because the assessment process changes. “Up five places” means very different things under those four scenarios.

Finally, a leaderboard should invite scrutiny rather than substitute for it. The productive use is to identify patterns and select debates for closer examination. The unproductive use is to cite a rank as if it settles the truth of a philosophical claim or a person's general intellectual worth.

12. What the next review should test

The next major GPT-model transition is the intended substantive review point for these frozen papers. A stronger ranking study would start with the scale calibration proposed in paper six: shared transcripts assessed under old and new procedures, with source fidelity and judgments reviewed independently.

It would then test prediction on genuinely new material. A ranking built from one set of debates should be evaluated against later, independently selected appearances, with the selection rules fixed before outcomes are known. Reusing the same scores in many random splits is useful for internal stability, but it is not the same as predicting a fresh sample.

Where possible, new collection should reduce the largest gaps in coverage. More appearances by scarcely observed speakers are valuable, but merely adding many similar debates by already frequent speakers will not solve uneven representation. Diversity of opponents, topics, roles, and formats matters alongside count.

The study should report whether conclusions survive reasonable alternative assumptions about ordinary variation. Some speakers may be more variable than others; some topics may be systematically harder to score; and recurring opponents can connect observations. No single model should be treated as an unquestionable machine for turning a selected archive into a universal ranking.

13. Conclusion: preserve the signal, retire the false precision

The ranking is not merely noise. The current repeatability estimate and split-record checks show a substantial, persistent speaker-associated pattern in the assessed archive. A reader can reasonably use the broad ordering as one guide to the performances the site has recorded.

But exact places make a more ambitious claim. A median neighboring gap of 0.17 points is tiny beside ordinary performance variation and the uncertainty created by small, uneven samples. Twelve-position resampling ranges and 19-position model ranges are not minor qualifications to a perfectly precise list. They change how the list should be read.

The responsible response is neither to abolish the leaderboard nor to defend every integer as a settled fact. **Keep the measured averages, show the amount of evidence behind them, and make the uncertainty part of the result.** A broad pattern can be useful without certifying who deserves tenth rather than eleventh place.

Most importantly, a rank is a summary of selected assessed performances - not a verdict on a person or a worldview. The leaderboard is at its best when it helps readers find arguments worth examining. It is at its worst when its apparent precision becomes another substitute for doing that examination.

Appendix: the complete 50-speaker field

The table is ordered by the published-sample mean, using appearance count and name to resolve display ties. Ranges cover the middle 95% of the corresponding simulations within this fixed field. The resampling column reuses each person's observed scores; the model column allows pooled ordinary variation. Neither interval includes all possible selection bias or judging error. The source scores are unchanged.

Speaker	Appearances	Mean	Resampled ranks	Model ranks
Joseph Schmid	6	87.33	1-5	1-9
Alex Malpass	6	87.17	1-12	1-9
Scott Clifton	4	86.75	1-10	1-12
Joe Folley	6	86.17	1.5-9	1-12
Sean Carroll	3	86.00	1-12	1-17
Matt Dillahunty	28	85.93	2-9	2-9
Alex O'Connor	28	85.46	3-10	2-10
Bart Ehrman	12	85.42	3-11	2-12
Lance Bush	5	84.80	2-16	2-18
Graham Oppy	12	84.08	7-15	5-16
Stephen Law	4	83.75	3-22.5	4-24
Daniel Dennett	6	83.17	7-23.5	7-24
Max Baker-Hytch	3	83.00	10-17.5	5-30
Michael Ruse	3	83.00	9-19	5-30
Michael Huemer	7	82.00	9-31	10-29
Bernardo Kastrup	5	81.80	14-24.5	10-31
Christopher Hitchens	11	81.73	13-25	12-28

Speaker	Appearances	Mean	Resampled ranks	Model ranks
Philip Goff	5	81.60	13-28	10-32
Richard Dawkins	7	81.43	11-32	11-31
Sam Harris	7	81.00	15-31	12-33
Michael Shermer	6	81.00	12-33	12-34
Stephen Woodford	6	81.00	16-30	12-34
Michael Licona	6	80.83	15-31.5	13-34
Ben Watkins	3	80.67	6-42	11-38
Josh Rasmussen	5	80.60	14-34	13-36
Richard Carrier	7	80.43	15-36	14-35
Lawrence Krauss	3	80.33	11.5-39.5	11-39
Robert Price	3	80.33	22-30	12-38
James Tour	4	80.25	20-31.5	13-38
William Lane Craig	24	79.50	23-35	23-36
David Wood	4	79.00	16.5-43	17-41
James White	3	79.00	28-36	15-42
Jonathan McLatchie	4	78.75	18-42	18-42
Tom Jump	5	78.00	29-41	23-43
Richard Swinburne	4	78.00	25-42	21-44
Ross Douthat	3	77.67	12-50	20-45
Glen Scrivener	5	77.60	28.5-43	25-44
John Lennox	9	77.11	31-43	31-44
Trent Horn	6	76.83	30-46	30-46
Aron Ra	4	76.75	33-45	27-47

Speaker	Appearances	Mean	Resampled ranks	Model ranks
Gregory Pine	3	76.00	33-46	28-48
Dinesh D'Souza	6	75.50	37-47	35-48
Keith Ward	3	75.33	29-50	31-49
Michael Jones	12	74.83	38.5-48	40-49
Jordan Peterson	6	74.83	40-47	37-49
Frank Turek	7	74.14	41-49	40-50
Alister McGrath	4	74.00	35-50	37-50
Peter Atkins	4	73.00	42-50	40-50
Matt Slick	3	72.33	46-50	40-50
Eric Hernandez	4	72.00	45.5-50	43-50

Methods and sources

The source revision is **76d006b37**, frozen September 4, 2026. Eligible appearances come from 237 comparable locked one-on-one assessments and use canonical site speaker identities. The default field requires at least three appearances, leaving 50 speakers and 334 observations. The entire eligible appearance set has 158 speakers and 474 observations.

The one-way repeatability model estimates between-speaker and within-speaker variance from unequal group sizes. Estimated repeatability for an average of n appearances is between variance divided by [between variance + within variance / n]. This describes the fitted observational sample under the model; it does not identify general ability or the correctness of the AI judgments.

Observed-score rank ranges use 20,000 independent within-speaker replacement resamples. A supplementary paired-weight analysis assigns one positive exponential random weight to each debate, shared by its two appearances, and re-ranks the fixed field. The pooled-variation model combines a normal distribution of persistent speaker levels with a common within-speaker variance; variance estimates are treated as fixed. Its 20,000 draws describe conditional uncertainty in latent means, not calibrated probabilities of future wins. Simulated ties receive average ranks.

Random split checks use 3,000 divisions among 23 speakers with at least six appearances. The sequence split orders by archive number, not recording date. Leave-one-out checks remove one appearance from one speaker at a time while holding the rest of the field fixed. Process-centering subtracts the relevant family mean before re-ranking and is a sensitivity exercise, not a causal correction.

The [analysis directory](#) provides the complete ranking data, formulas, simulations, source hashes, and figure contracts. The [Backend library](#) links the six companion papers, especially the scale audit on which any stronger cross-debate interpretation depends. No public assessment or official speaker score is altered by this paper.